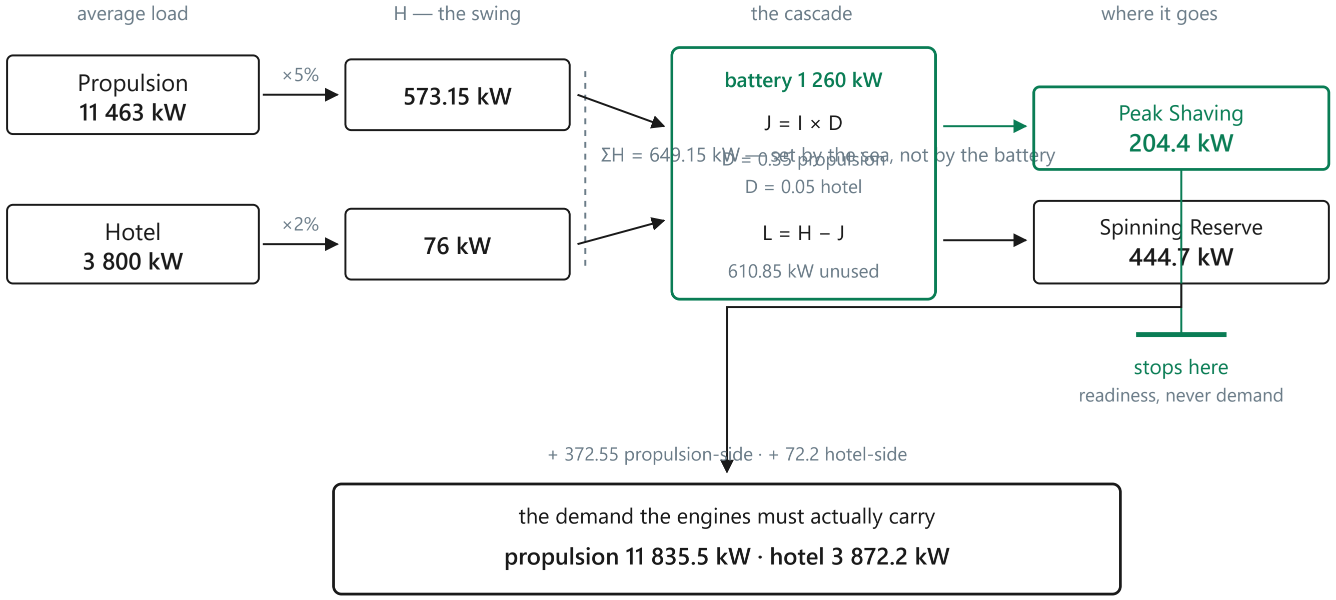
00a — The pipeline, drawn

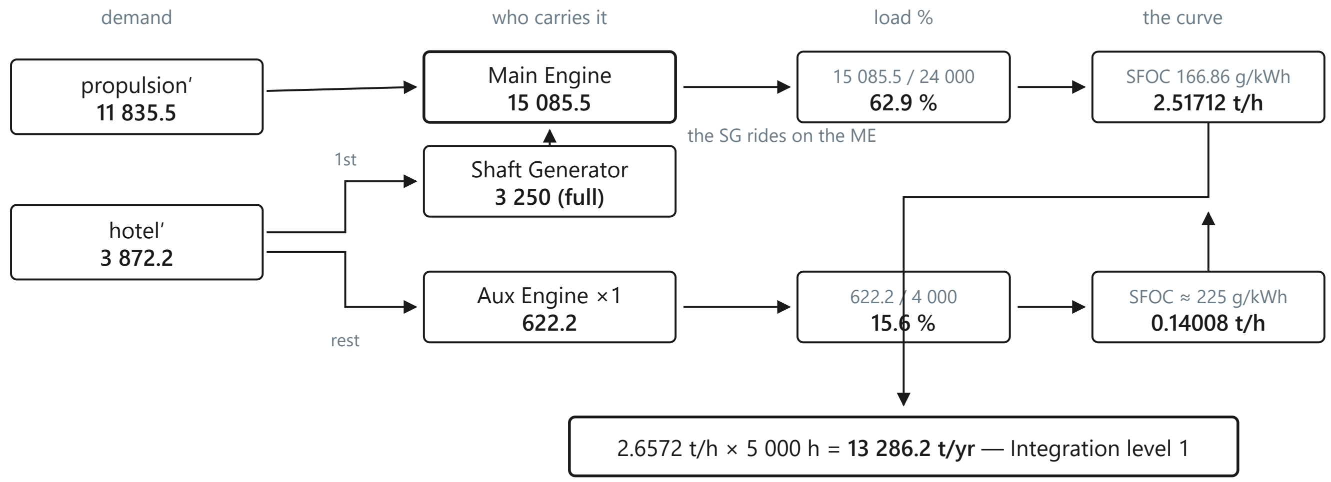
Three pictures, three claims. All numbers are scenario **01** (propulsion 11 463 kW · hotel 3 800 kW · battery 1 260 kW · Transit 5 000 h · MDO at $780/t), so every figure here can be checked against `01-excel-baseline.pdf` line by line.

1 — The battery does not reduce the swing. It decides who carries it.



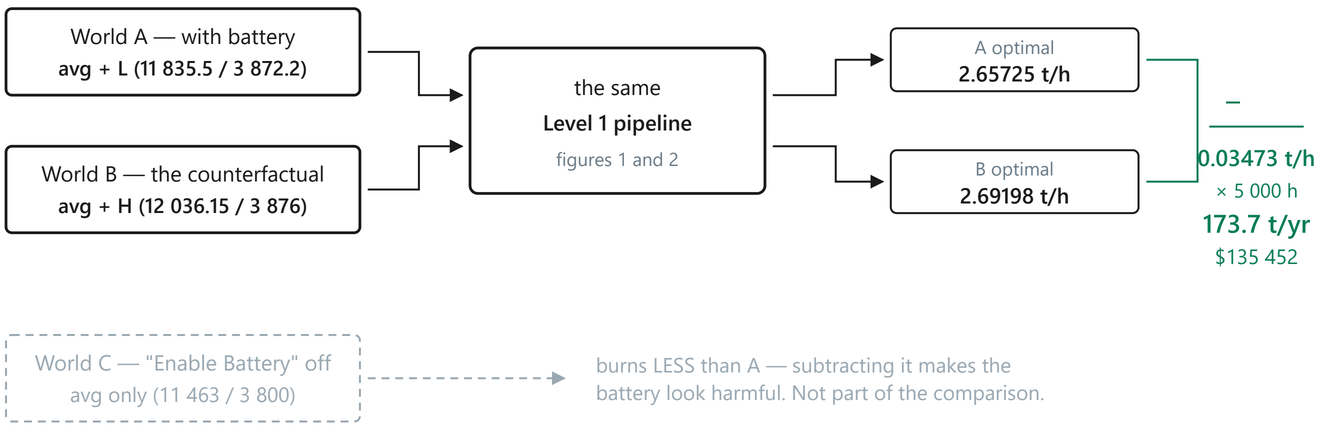
> **ΣJ + ΣL = ΣH, always.** A bigger battery moves the split; it never shrinks the 649.15. Everything the battery covers (green) stops at the tile — a covered reserve is readiness, not load, so it is never subtracted from anything. Everything it misses is added to the demand instead. This is why scenario 02, with a 300 kW battery, shows exactly the same ΣH but PS 105 / SR 544.15.

2 — Demand becomes fuel through the load split, then the curve



> **Two rules do all the work here.** The shaft generator is filled before any auxiliary starts, because the main engine is already turning; and the SG's output is a load *on* that main engine, which is why the ME figure exceeds propulsion. The curve does the rest: the same 622 kW spread over three auxiliaries instead of one puts each at 5.2 % where the engine burns ~232 g/kWh instead of ~225 — and that difference, multiplied by 5 000 hours, *is* the Level 1 saving.

3 — The Battery Benefit is the same pipeline, run twice



> **Both figures compared are optima, never baselines** — a pinned baseline is deliberately ignored in the counterfactual, or the comparison would stop being like-for-like. **World B is scenario 03**, written down as its own file: its inputs are 01's raw loads plus the full swing, which is why 03's Level 1 (13 459.9 t/yr) minus 01's (13 286.2) reproduces the badge exactly. To check a Benefit by hand, load 03 — do not untick the battery.

A number you can carry in your head

Figure 1 hands the engines **204.4 kW less** to carry; figure 2 says every kilowatt costs roughly its SFOC in fuel. So:

`Benefit ≈ Peak Shaving × SFOC × hours`

204.4 kW × 167 g/kWh × 5 000 h ÷ 10^6 = **170.7 t/yr** against the true 173.7

The remaining ~3 tonnes come from the engines settling at slightly better points on the curve. The approximation is close enough to sanity-check any scenario in seconds — and it shows why `CoverageFactor` is the most consequential setting in `appsettings.json`: it alone decides how much of the swing ever becomes Peak Shaving.